A1.1 Water

NAME	DATE	TOTAL
		_____ / 36

About 35 minutes, or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make.

Section A: Recall

7 MARKS

1 STATE the type of bond between atoms within a water molecule. [A1.1.2] [1]

2 STATE the type of attraction between adjacent water molecules. [A1.1.2] [1]

3 DEFINE adhesion. [A1.1.4] [1]

4 DEFINE hydrophilic. [A1.1.5] [1]

5 LIST three physical properties of water that affect animals living in water. [A1.1.6] [3]

Section B: Understanding

13 MARKS

6 DRAW two water molecules, with a hydrogen bond between them, showing polarity. [A1.1.2] [3]

7 EXPLAIN how cohesion helps water move up a tall tree. [A1.1.3] [3]

8 **EXPLAIN** why glucose is transported dissolved in blood plasma, but cholesterol is not. [A1.1.5] [4]

9 **OUTLINE** why the hydrophobic nature of some molecules is useful to cells. [A1.1.5] [3]

Section C: Working with data

10 MARKS

Values in question 10 are approximate. Results in question 11 are invented for practice.

10 [A1.1.6]

PROPERTY, AT 20°C	WATER	AIR
DENSITY / KG M^{-3}	998	1.2
VISCOSITY / MPA S	1.0	0.018
THERMAL CONDUCTIVITY / W M^{-1} K^{-1}	0.60	0.026

a **CALCULATE** how many times greater the thermal conductivity of water is than that of air. [1]

b **EXPLAIN** how this difference affects a seal that moves between water and air. [2]

c **EXPLAIN** how the difference in density affects a bird that both flies and dives. [2]

11 [A1.1.3, A1.1.4]

Students counted how many drops of liquid fit on a coin before spilling.

LIQUID	TRIAL 1	TRIAL 2	TRIAL 3
WATER	34	38	36
COOKING OIL	14	12	13

a **CALCULATE** the mean number of drops for each liquid. [2]

b **EXPLAIN** the difference in terms of molecular attraction. [3]

Section D: Longer answer

6 MARKS

12 **EXPLAIN** how hydrogen bonds give water properties that are useful to living organisms. [A1.1.2–6] [6]